

Appendix – Supplementary materials

Table S1. Related parameters of ultrasonic instrument and probe.

Ultrasonic instrument, Producer	Contrast imaging mode	Probe, Frequency	Mechanical index
Acuson Sequoia 512®, Siemens, Mountain View, Calif.	Contrast Pulse Sequencing® (CPS)	Vector array® (model 4V1), 1-4MHz	0.07-0.19
Aloka SSD-Alpha 10®, Aloka, Japan	Contrast Harmonic Echo® (CHE)	Convex array® (model UST-9130), 2-6MHz	0.06-0.10
Toshiba Aplio XV/500®, Toshiba, Japan	Contrast Harmonic Imaging® (CHI)	Convex array® (model PVT375BT), 3.5MHz	0.07-0.12
Aixplorer, SuperSonic Imagine, Aix-en-Provence, France	Pulse inversion-power modulation techniques	Curved-array transducer (model SC6-1), 1.0–6.0MHz	0.05–0.13
Canon Aplio i900, Canon, Japan	Contrast Harmonic Imaging® (CHI)	Convex array® (model PVT375BT), 3.5MHz	0.07-0.12

Table S2. Performance evaluation of Rad models.

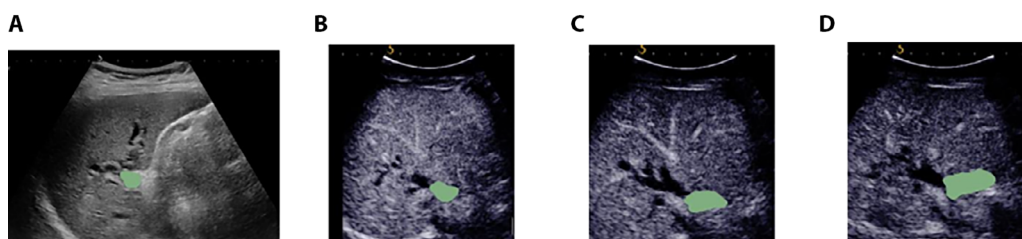
Training group				
Model	AUC (95%CI)	Accuracy (95%CI)	Sensitivity (95%CI)	Specificity (95%CI)
US-Rad	0.84 (0.75 - 0.93)	0.79 (0.69 - 0.87)	0.74 (0.60 - 0.89)	0.83 (0.70 - 0.93)
CEUS-Rad	0.88 (0.81 - 0.95)	0.80 (0.71 - 0.89)	0.86 (0.74 - 0.97)	0.75 (0.63 - 0.88)
Combination-Rad	0.88 (0.80 - 0.95)	0.81 (0.73 - 0.89)	0.94 (0.86 - 1.00)	0.70 (0.55 - 0.83)
Internal validation group				
Model	AUC (95%CI)	Accuracy (95%CI)	Sensitivity (95%CI)	Specificity (95%CI)
US-Rad	0.64 (0.40 - 0.89)	0.60 (0.40 - 0.76)	0.88 (0.63 - 1.00)	0.47 (0.24 - 0.71)
CEUS-Rad	0.66 (0.58 - 0.93)	0.72 (0.52 - 0.88)	0.63 (0.55 - 0.88)	0.76 (0.53 - 0.94)
Combination-Rad	0.70 (0.42 - 0.98)	0.72 (0.52 - 0.88)	0.75 (0.58 - 1.00)	0.71 (0.47 - 0.94)
External validation group				
Model	AUC (95%CI)	Accuracy (95%CI)	Sensitivity (95%CI)	Specificity (95%CI)
US-Rad	0.56 (0.15 - 0.97)	0.60 (0.30 - 0.90)	0.80 (0.40 - 1.00)	0.40 (0.30 - 0.80)
CEUS-Rad	0.64 (0.23 - 1.00)	0.70 (0.50 - 0.90)	0.80 (0.40 - 1.00)	0.60 (0.20 - 0.80)
Combination-Rad	0.68 (0.27 - 1.00)	0.80 (0.60 - 1.00)	1.00 (0.30 - 1.00)	0.60 (0.20 - 1.00)

Table S3. DLR models Performance Based on Repeated Cross-validation.

Model	Average AUC (95%CI)	Average Accuracy (95%CI)	Average Sensitivity (95%CI)	Average Specificity (95%CI)
BUS-DLR	0.84 (0.79 - 0.88)	0.87 (0.84 - 0.90)	0.90 (0.85 - 0.94)	0.85 (0.80 - 0.89)
CEUS-DLR	0.88 (0.85 - 0.92)	0.89 (0.85 - 0.92)	0.94 (0.90 - 0.98)	0.84 (0.79 - 0.89)
Combination-DLR	0.93 (0.90 - 0.97)	0.94 (0.92 - 0.97)	0.97 (0.95 - 1.00)	0.91 (0.87 - 0.95)

Table S4: Performance evaluation of DLR models.

Training group				
Model	AUC (95%CI)	Accuracy (95%CI)	Sensitivity (95%CI)	Specificity (95%CI)
BUS-DLR	0.97 (0.93 - 1.00)	0.93 (0.88 - 0.99)	1.00 (0.99 - 1.00)	0.88 (0.78 - 0.98)
CEUS-DLR	0.99 (0.97 - 1.00)	0.97 (0.93 - 1.00)	0.97 (0.91 - 1.00)	0.98 (0.93 - 1.00)
Combination-DLR	0.94 (0.89 - 0.99)	0.87 (0.79 - 0.93)	0.91 (0.83 - 1.00)	0.83 (0.70 - 0.93)
Internal validation group				
Model	AUC (95%CI)	Accuracy (95%CI)	Sensitivity (95%CI)	Specificity (95%CI)
BUS-DLR	0.70 (0.57 - 0.93)	0.72 (0.52 - 0.88)	0.63 (0.55 - 1.00)	0.76 (0.53 - 0.94)
CEUS-DLR	0.77 (0.52 - 0.98)	0.80 (0.64 - 0.96)	0.75 (0.58 - 1.00)	0.82 (0.65 - 1.00)
Combination-DLR	0.83 (0.65 - 0.95)	0.88 (0.76 - 0.99)	0.88 (0.63 - 1.00)	0.88 (0.71 - 1.00)
External validation group				
Model	AUC (95%CI)	Accuracy (95%CI)	Sensitivity (95%CI)	Specificity (95%CI)
BUS-DLR	0.66 (0.26 - 1.00)	0.70 (0.40 - 1.00)	0.80 (0.40 - 1.00)	0.60 (0.20 - 1.00)
CEUS-DLR	0.68 (0.28 - 1.00)	0.70 (0.40 - 0.90)	0.80 (0.40 - 1.00)	0.60 (0.20 - 1.00)
Combination-DLR	0.72 (0.34 - 1.00)	0.80 (0.60 - 1.00)	1.00 (0.50 - 1.000)	0.60 (0.20 - 1.00)

**Figure S1:** This figure illustrates the ROIs for HCCA tumor lesions across four phases: A) BUS, B) peak enhancement CEUS, C) venous phase CEUS, and D) delayed phase CEUS.

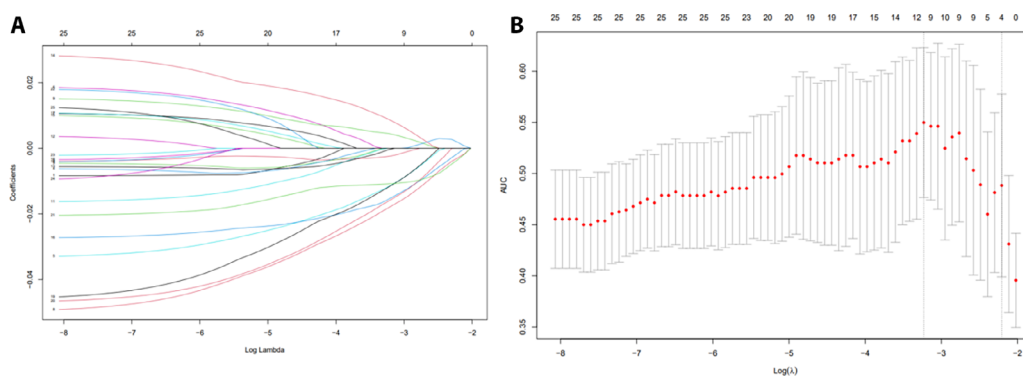


Figure S2. Screening of BUS-RadFs. A) The feature coefficient decreases with the increase of λ value. B) The optimum λ value is selected by 10-fold cross-validation.

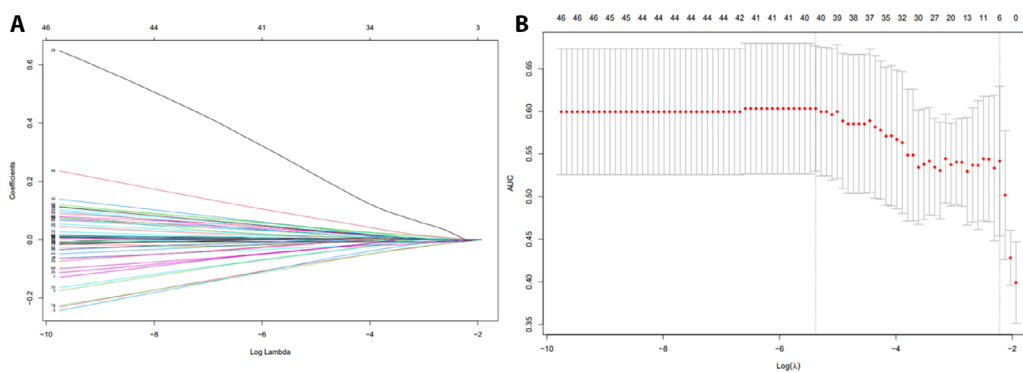


Figure S3: Screening of CEUS-RadFs. A: the feature coefficient decreases with the increase of λ value. B: The optimum λ value is selected by 10-fold cross-validation.

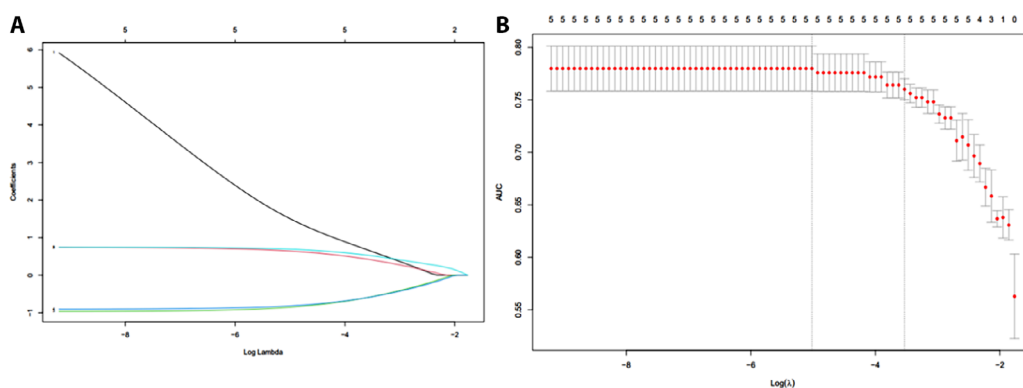


Figure S4. Screening of BUS-DLFs. A) The feature coefficient decreases with the increase of λ value. B) The optimum λ value is selected by 10-fold cross-validation.

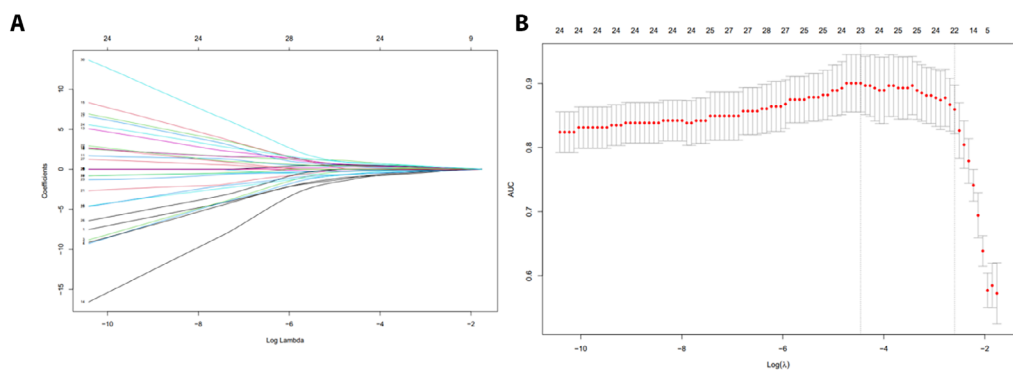


Figure S5. Screening of CEUS-DLFs. A) The feature coefficient decreases with the increase of λ value. B) The optimum λ value is selected by 10-fold cross-validation